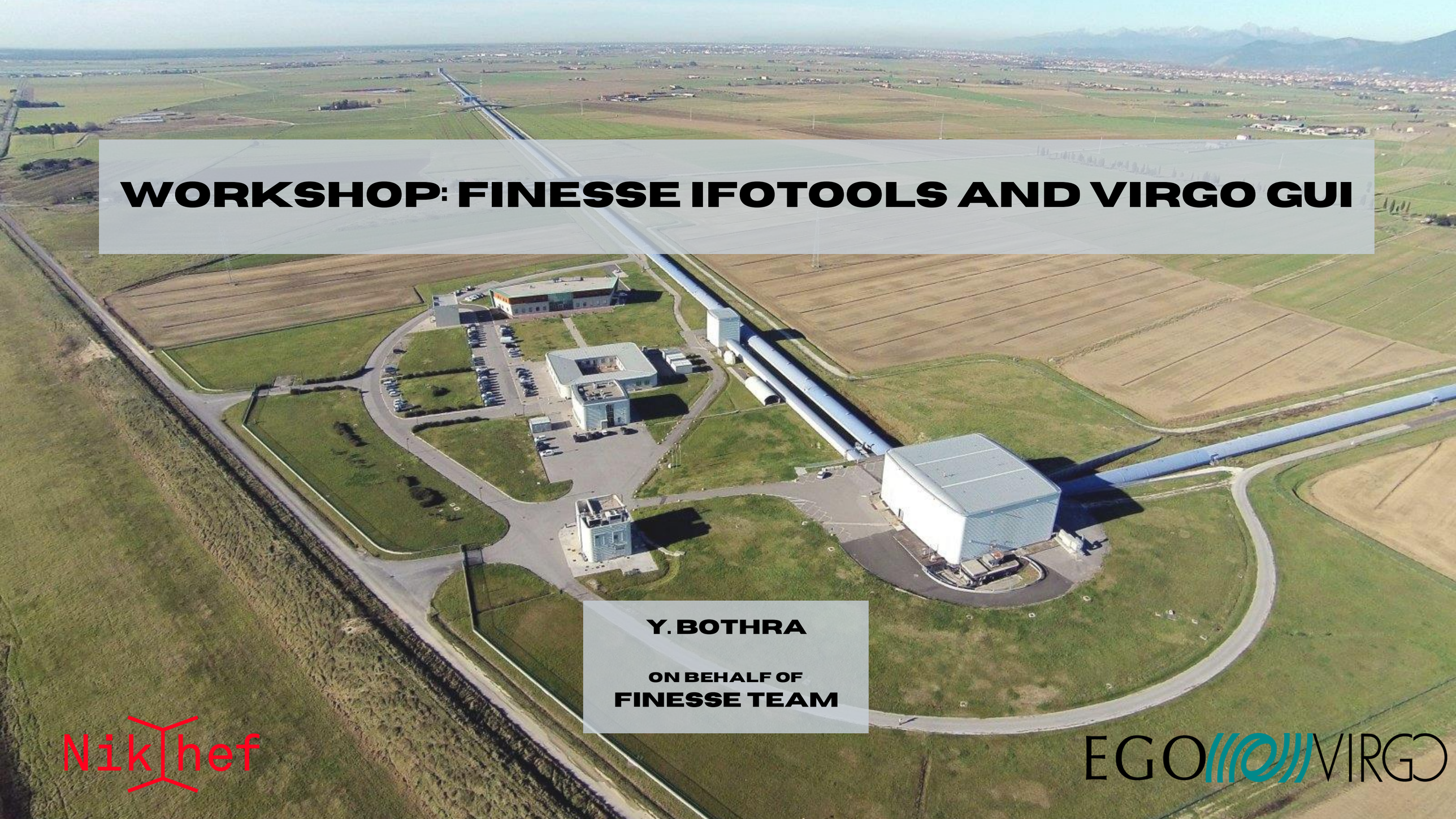
WORKSHOP: FINESSE IFOTOOLS AND VIRGO GUI



Y. BOTHRA

ON BEHALF OF
FINESSE TEAM

Nikhef

EGO VIRGO

Contents

- Introduction
- Agenda
- First session
- Links to relevant material

Introduction: why are we here?

IFOTOOLS

To present Ifotools and Virgo GUI

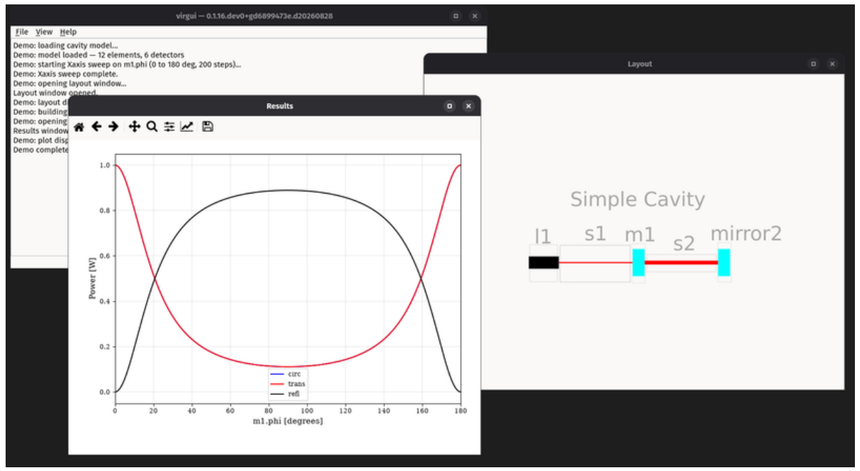
- A. Set of simple python functions.
- B. Designed to simplify simulations, easy to debug and code review, sets the right method /steps for simulations.
- C. Suggest modeling structure, and methods.

1 ifotools package
1.1 Subpackages
1.1.1 ifotools.finesse package
1.1.2 ifotools.mirror_maps package
1.1.3 ifotools.virgo package
1.2 Submodules
1.2.1 ifotools.fft_propagation module
1.2.2 ifotools.grating module
1.2.3 ifotools.helper module
1.2.4 ifotools.sympy_helpers module
1.2.5 ifotools.utils module
1.2.6 ifotools.version module
1.3 Module contents
1.3.1 Parameters
1.3.2 Examples

More on this later today…

GUI

- User interface with pre-defined tools - designed to perform quick simulations.
- Visualization: plots, detector responses, optical model
- Currently in early design phase.



Introduction: how do we do this?

Notebooks:

- To showcase the use of ifotools
- Example notebooks: Pretuning, lock acquisition, O5 sensing and control.
- Perform simulations / dedicated study using these tools.

Cross-collaboration:

- Discuss possible ways to maintain this repository and certain sub-packages.
- Brainstorm more tools/functions & GUI that can be needed or necessary.
- How do we do this in the long run..?

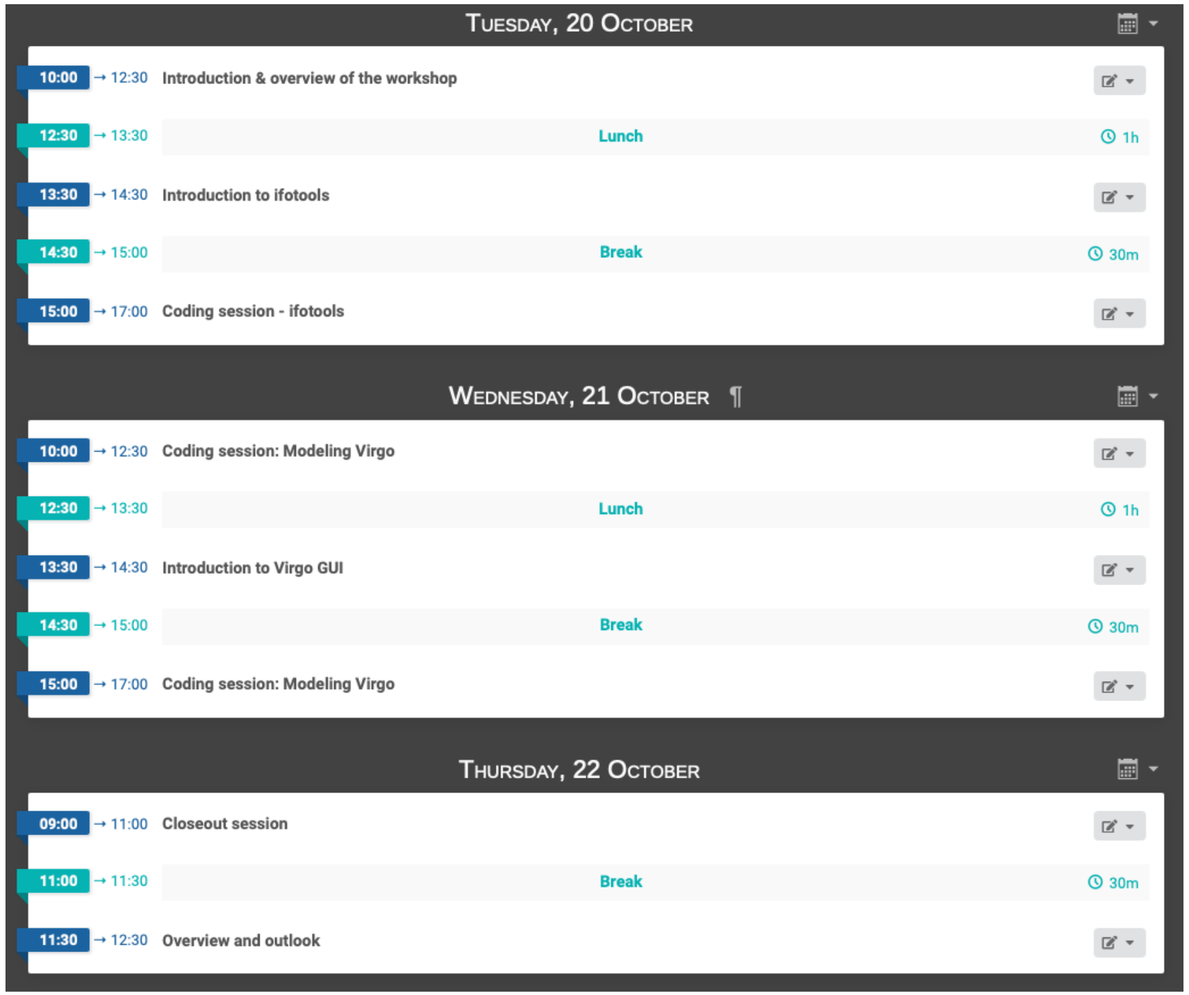
Agenda

FIRST SESSION

INTRODUCTION TO IFOTOOLS AND EXAMPLE NOTEBOOKS

CODING SESSIONS USING IFOTOOLS

WRAP-UP AND CLOSEOUT SESSION



Tuesday, 20 October

10:00 → 12:30	Introduction & overview of the workshop	
12:30 → 13:30	Lunch	1h
13:30 → 14:30	Introduction to ifotools	
14:30 → 15:00	Break	30m
15:00 → 17:00	Coding session - ifotools	

Wednesday, 21 October

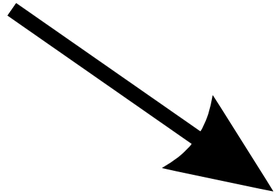
10:00 → 12:30	Coding session: Modeling Virgo	
12:30 → 13:30	Lunch	1h
13:30 → 14:30	Introduction to Virgo GUI	
14:30 → 15:00	Break	30m
15:00 → 17:00	Coding session: Modeling Virgo	

Thursday, 22 October

09:00 → 11:00	Closeout session	
11:00 → 11:30	Break	30m
11:30 → 12:30	Overview and outlook	

First Session

- Formal introduction: name, work/affiliation, specialization, how do you use finesse?, hobbies, fun facts, etc.
- Introduction to Finesse → **Jeremie**
- Brainstorming for coding session → **for tomorrow**



Notebooks:

- To showcase the use of ifotools
- Example notebooks: Pretuning, lock acquisition, and O5 sensing and control.
- Perform simulations / dedicated study using these tools.

Links to relevant material and repository

- Finesse 3 documentation: https://finesse.ifosim.org/docs/latest/
- Working repository for the workshop: https://gitlab.com/ifosim/finesse/finesse3_workshop
- Ifotools repository: https://gitlab.com/ifosim/ifotools
- Presentations and other relevant documents: https://gitlab.com/ifosim/finesse/finesse3_workshop/-/tree/main/documents?ref_type=heads
- Minutes of the workshop: https://docs.google.com/document/d/1_pOzNI1Kqzty7egEAc6hsl-FRc2790bdDhinyYhxi_8/edit?usp=sharing